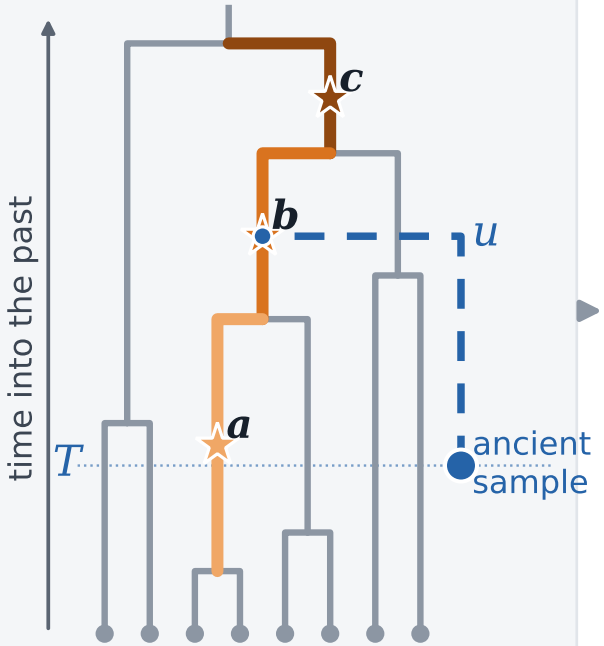
Dating ancient genomes by threading them into a modern ARG

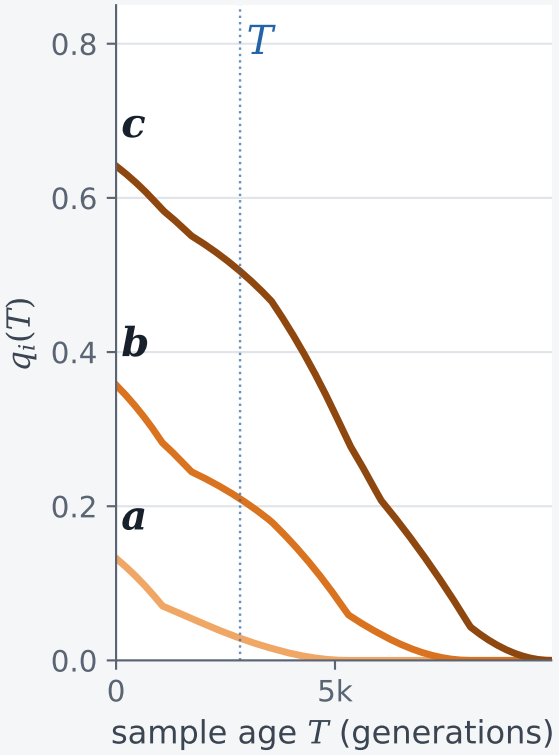
1 Insert the lineage



$$S_T(u) = \exp\left[-\int_T^u \frac{\kappa(s)}{2N_e(s)}\,ds\right]$$

κ: tree lineages at time s
$S_T(u)$: not yet joined by time u

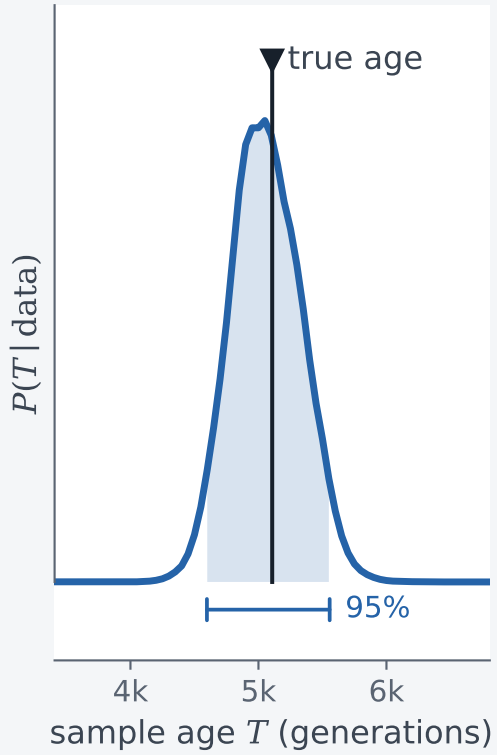
2 One curve per site



$$q_i(T) = \int_T^{z_i} S_T(u)\,\frac{\delta_i(u)}{2N_e(u)}\,du$$

δ_i: lineages below mutation i
z_i: mutation time (averaged over edge)

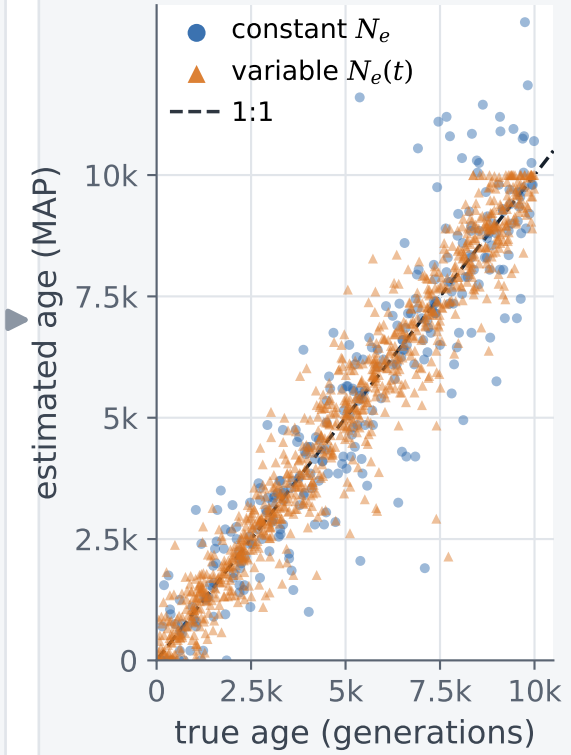
3 Combine sites



$$L(T) = \prod_i q_i(T)^{x_i}\,[1 - q_i(T)]^{1 - x_i}$$

$x_i = 1$: derived allele observed
one simulated sample, ~40k sites,
true modern ARG

4 Simulation check



$$\hat{T} = \arg\max_T P(T \mid \text{data})$$

true ARG; mean $\hat{T} - T$, slope
constant N_e: +13 gen, 1.00
variable $N_e(t)$: +8 gen, 0.99